The Child Health Burden of Unobserved Water*

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Abstract

How much disease is caused by environmental hazards that existing observation systems fail to detect? One such hazard is transient stagnant water, which can transmit enteric pathogens and provide mosquito breeding habitat but remains largely invisible to conventional satellite monitoring[2,3]. Here we develop a 90-m hydrological model to reconstruct stagnant water across Tanzania, validate it against independent wetland and soil-moisture observations, and estimate its health effects using nationally representative child-health data. We find that moving from zero to 3% of surrounding land under stagnant water increases two-week diarrhoea prevalence by 4.8 percentage points (95% CI, 1.5–8.2), equivalent to 45% of mean prevalence, and reduces haemoglobin by 2.41 g/L (95% CI, 0.3–4.6). The diarrhoeal response is substantially larger among households relying on surface drinking water, whereas haemoglobin reductions are concentrated in malaria-suitable areas, consistent with complementary enteric and vector-borne pathways[1,4,5]. We estimate that stagnant water accounts for 11.3% (95% CI, 4.3–18.7%) of under-five diarrhoea in Tanzania, corresponding to approximately 445 child deaths annually from the diarrhoeal pathway alone. In a policy simulation, targeting a one-percentage-point expansion in national oral-rehydration coverage towards high-exposure areas would avert an estimated 139 under-five deaths annually, twice as many as untargeted allocation. Because the model inputs are globally available, this framework can be extended beyond Tanzania to map previously unobserved stagnant-water exposure and improve the targeting of health interventions.

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Introduction

Unsafe water remains a major threat to child health, particularly in low-income settings[1], despite substantial investments in drinking water, sanitation and hygiene[6,7]. One especially difficult-to-measure hazard is routine stagnant water: shallow, transient pools that remain for days or weeks in low-lying terrain after rainfall or flooding (Fig. 1a). These pools can transmit enteric pathogens through contaminated water[8], provide breeding habitat for *Anopheles* mosquitoes[9–11], and support water-contact parasites such as schistosomes[12]. Local studies have linked surface-water dynamics to diarrhoeal disease[13,14], while field and process-based studies show that fine-scale hydrology determines mosquito larval habitat[10,15]. Despite these established mechanisms, stagnant water has not been measured consistently at population scale in a way that can be linked to health and used to identify where and when exposure is greatest.

A central challenge is measurement. Global optical surface-water products have transformed the mapping of permanent and seasonal open water[16], but shallow transient pools are difficult to observe because they are small, short-lived, shallow, turbid and often vegetated[2,3,17]. This matters for health: higher-resolution surface-water measures outperform both precipitation and widely used global optical surface-water products in predicting malaria risk[18]. Optical products observe only water that is visible when the surface is viewed, missing pools that form and drain between observations or remain obscured by clouds or vegetation.

To overcome this measurement gap, we develop a process-based framework for mapping transient stagnant water at high spatial and temporal resolution and apply it nationwide in Tanzania. The model combines rainfall, evaporation, soil infiltration and 90-m topography in a water-balance and hydraulic-routing system that determines where and when water accumulates across the landscape. Tanzania provides a national-scale setting spanning substantial variation in hydrology, malaria ecology and household water access, together with nationally representative child-health data. The resulting exposure field resolves the formation and persistence of stagnant water at sub-daily resolution, can be validated against independent observations of wetlands and surface wetness, and can be linked directly to recent health-exposure windows. Because the required meteorological, soil and topographic inputs are globally available, the framework is portable beyond Tanzania.

This process-based approach also provides an inferential advantage over directly observed surface water. The ideal experiment would expose otherwise similar nearby communities to different amounts of stagnant water under the same weather and broader environmental conditions. Our model approximates this design by exploiting how common rainfall is transformed by highly local terrain, soils and drainage into different levels of ponding across nearby locations. We then compare children within the same local area and survey wave, absorbing common weather, seasonality and local shocks. The identifying variation therefore comes from fine-scale differences in where water accumulates, rather than from broad differences in rainfall itself.

A large literature uses rainfall and other meteorological variables to estimate population health effects, including effects of precipitation variability and extreme rainfall on childhood infectious disease and mortality [19–22] and of temperature and precipitation on childhood malaria across Africa [23]. But rainfall is an upstream forcing that affects health through multiple channels [24] and does not uniquely determine where water accumulates. Recent work has therefore moved toward process-based measures of water availability, which can materially alter estimates of malaria suitability [25,26]. Current large-scale approaches, however, typically operate at spatial scales too coarse to resolve the shallow, transient pools relevant for enteric and mosquito exposure [25]. The approach taken here increases resolution by close to three orders of magnitude.

We show that this previously unobserved water has measurable and pathway-specific consequences for child health. Stagnant-water exposure increases childhood diarrhoea, with substantially larger effects among households relying on surface water, and lowers haemoglobin in malaria-suitable areas, with corresponding increases in fever and anaemia. These two responses dissociate along the mechanisms predicted by enteric and vector transmission. The resulting diarrhoeal burden is highly concentrated across space and seasons, allowing the exposure map to identify where additional health coverage has the greatest potential impact. As one concrete demonstration, using stagnant-water predictions to target a fixed expansion in oral-rehydration coverage approximately doubles the number of child deaths averted relative to untargeted allocation.

More generally, representing stagnant water as a physical environmental state rather than an upstream weather proxy creates a link between climate, exposure and health that can be updated with recent observations, driven by weather forecasts, or run under alternative climate conditions. This is particularly relevant as climate change intensifies heavy precipitation and alters the hydrological cycle [27], while a substantial share of the welfare costs of climate change is expected to operate through human health, an impact that is still poorly understood in low-income country settings [28]. Because the required inputs are globally available and harmonized [29–32], the same framework can in principle be extended across countries, climates and time periods.

Mapping transient stagnant water at national scale

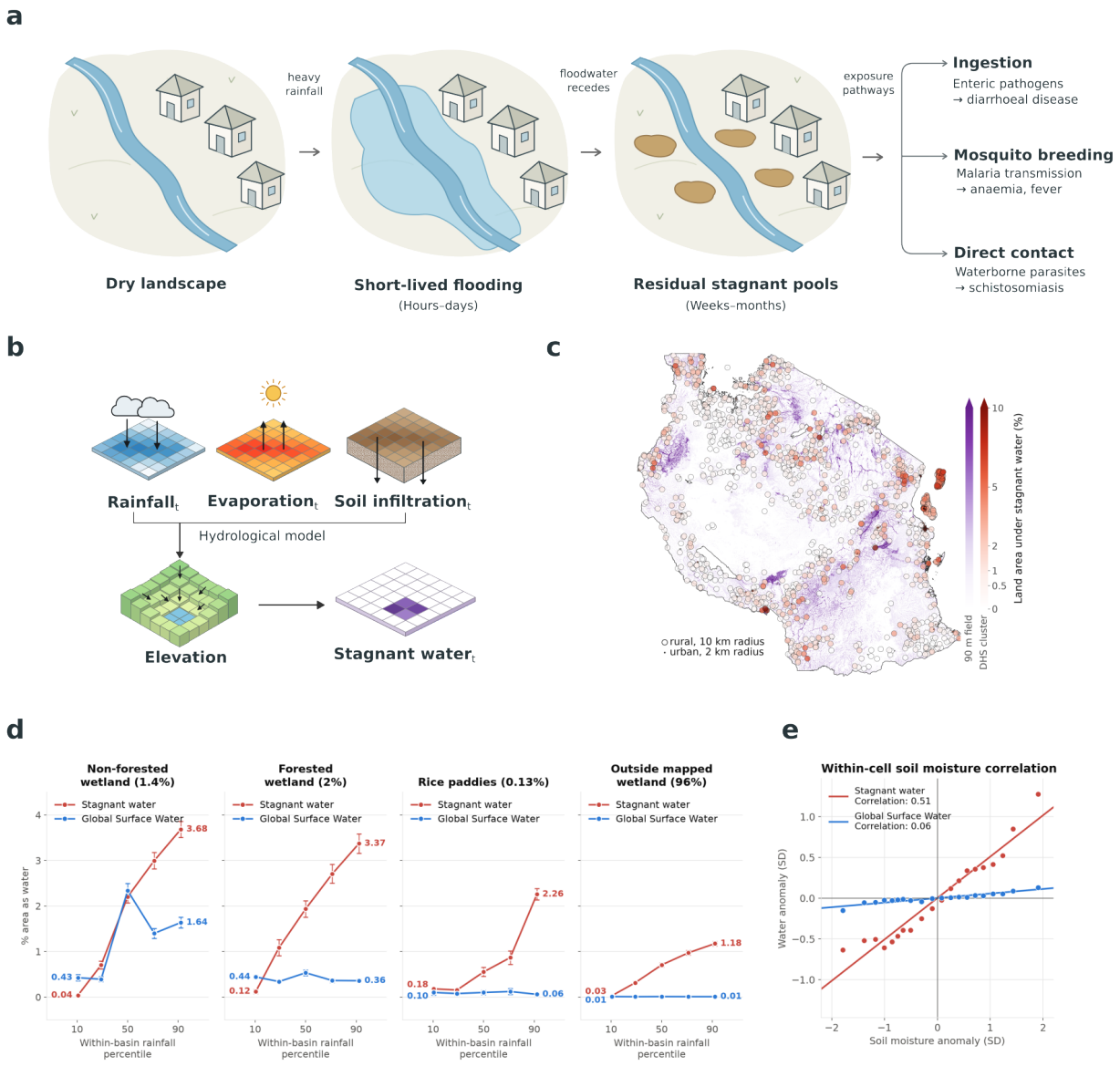


Fig. 1: Exposure pathways of stagnant water pools and how they are mapped. a, Short-term floods and heavy rainfall generate stagnant water pools that may last for weeks and trigger multiple disease pathways. **b,** The stagnant water model combines gridded climatological and topographic inputs to simulate surface water generation over time at a 90 m resolution. **c,** Circles shaded in red show 8-week average stagnant water exposure around each DHS survey cluster, while long-term average stagnant water prevalence is shown in shades of purple. **d,** The stagnant water model classifies more mapped wetlands (GLWD v2[33]) than an optical satellite product (Global Surface Water[16]) and exhibits a more consistent seasonal pattern. **e,** Predicted stagnant water anomalies correlate more strongly than Global Surface Water with state-of-the-art soil moisture data that is less sensitive to cloud and vegetation cover, as well as shallow depths and brown coloration.

We model transient stagnant water as the residual of an explicit water balance solved on a 90 m grid, using globally harmonized gridded data on elevation, rainfall, evaporation and soil infiltration (Fig. 1b; Methods). These inputs represent the current state of the art for hydrological modelling in data-scarce settings such as Tanzania; in particular, MSWEP is among the best-performing global precipitation datasets in gauge-based and hydrological-model evaluations[31,34]. Rainfall enters from three-hourly MSWEP fields, disaggregated into 5-minute time steps. Infiltration is removed at each time step using the local saturated hydraulic conductivity of the soil[30], since most of the infiltration loss occurs during the ponding stage, when the soil layer underneath is assumed to reach near-saturated conditions. Water that remains on the surface is redistributed across the 90 m SRTM digital elevation model[29] by a local mass-balancing hydraulic rule: water is routed from each cell to its nearest neighbor depending on steepest hydraulic gradient at each time step. Potential evapotranspiration rates from ERA5[32] are then applied on cells with a positive water depth at an hourly level.

The scheme is mass-conserving, strictly local, and has no free parameters beyond the physical inputs. Water accumulates in low-lying areas where the landscape gives it nowhere to go, which is the object of interest. Built-up land, where no infiltration occurs, is assigned a drainage capacity representing low-capacity storm-water infrastructure. The resulting standing-water outputs are a coarse representation of the local surface-water balance and abstract from subsurface processes such as groundwater and soil-water flows into rivers and springs.

The run time of hydrodynamic simulations is highly sensitive to spatial resolution, typically scaling with its cube. However, evaluations of large-scale 2D hydrodynamic simulations for several European rivers indicate that flooded areas and water levels change little once the grid is finer than about 100 m[35–37]. This makes it feasible to simulate the formation of stagnant water across a whole country such as Tanzania over time, and our 90 m grid is consistent with this guideline. The simplified routing is not, however, an accurate description of short-term river flood dynamics, which would require river cross-section data and full hydraulic solutions that are infeasible at national scale, over months to years of simulation, and given the data scarcity of a low-income country such as Tanzania. The model is written in Python and will be made available at the time of publication.

The model is run over the full set of windows spanning the 1999, 2010, 2016 and 2022 georeferenced Demographic Health Surveys (DHS)[38] waves available for Tanzania, extending back at a minimum twelve weeks before the first interview date in each survey wave. This is done in order to initialize the model, and provide sufficient results to produce 8-week average exposure before each interview, while reducing computational time.

We define a cell as covered by stagnant water if it has depth of at least 5 cm and water is standing still in the cell (no transport in or out). We then link stagnant water exposure to the child health data by computing the share of land under water in a 2 km (urban) or 10 km (rural) buffer around each survey cluster coordinate, averaged over the eight weeks preceding interview, using a triangular

kernel (Methods; Fig. 1c).

We validate the model outputs against three external datasets (Fig. 1d–e). We begin with the most direct benchmark, JRC Global Surface Water (GSW), which maps surface-water occurrence from Landsat imagery back to 1984. The within-grid-cell correlation between modelled stagnant water and GSW is surprisingly low (+0.006). There are two possible interpretations. One is that the model is poorly capturing actual surface water. The other is that GSW is itself a weak benchmark for the type of water we aim to measure: transient, shallow water that often appears during the wet season and in vegetated environments, both of which are comparatively difficult to observe with optical satellite imagery because of cloud cover and vegetation occlusion. Since our object of interest is transient stagnant water, we exclude permanent water bodies from this comparison, removing precisely the settings in which optical surface-water products perform most strongly.

Wet-season imagery from May 2022 around the village of Muchirobi illustrates this measurement gap (Extended Data Fig. 1). Modelled stagnant-water cells overlap portions of the visible flood footprint that lie beyond the area marked as ever observed water in the optical product. This local spatial comparison complements the national tests of seasonal and month-to-month variation below.

To assess these patterns systematically, we next compare both measures with the Global Lakes and Wetlands Database v2 (GLWD v2), which maps wetlands globally at approximately 500 m resolution[33]. The model identifies substantially more water than GSW within every mapped wetland class and, importantly, responds much more strongly to the wet season. In riverine floodplains, modelled stagnant-water coverage rises from 1.5% of area in dry basin-months to 3.3% in wet basin-months, while GSW increases only from 0.4% to 0.5%. In forested wetlands, modelled coverage rises from 1.4% to 3.2%, whereas GSW shows essentially no seasonal response. The strongest seasonal response in GSW occurs in non-forested wetlands, where optical detection is comparatively less obstructed by vegetation. Even outside areas classified as wetlands by GLWD, the model predicts approximately 1.0% of area to be covered by stagnant water during wet months, while GSW detects almost none (Fig. 1d). These patterns suggest that the weak correspondence with GSW is concentrated in precisely the environments in which optical surface-water detection is most difficult, while the model places water in hydrologically plausible locations and responds to seasonal rainfall in the expected direction. An important limitation is that GLWD is static and therefore cannot validate short-run temporal variation directly.

As a third and final validation exercise, we compare the stagnant-water measure with NASA's Soil Moisture Active Passive (SMAP) satellite observations[39]. SMAP uses L-band microwave radiometry to measure near-surface soil moisture, providing an independent measure of surface wetness that, unlike optical remote-sensing products, is largely unaffected by cloud cover and does not rely on visible surface reflectance. We exploit monthly within-cell variation and regress standardized soil-moisture anomalies on standardized anomalies in each water measure. A one-standard-deviation increase in modelled stagnant water is associated with a 0.46-standard-deviation

increase in SMAP soil moisture (0.47 using the assimilated Level-4 product), compared with only 0.05 standard deviations for the satellite surface-water product. Thus, the stagnant-water model not only identifies water in locations and seasons where it is expected to occur most frequently, such as wetlands during the wet season, but also closely tracks independent month-to-month variation in surface wetness within locations.

Stagnant water increases childhood diarrhoea

We link stagnant-water exposure to 32,254 children under five across four Tanzania Demographic and Health Survey waves[38]. Identification comes from comparing children in different survey clusters within the same 0.5° cell and survey wave, net of cell-specific seasonality, cell-by-wave shocks, rainfall, and household and child characteristics (Methods). We summarize the dose–response at 3% of surrounding land under stagnant water, approximately the 93rd percentile of exposure. This level captures the high-exposure tail in which the burden is concentrated and on which the targeting exercises below operate. All dose–response curves are normalized to zero at zero stagnant-water exposure.

Stagnant water substantially increases childhood diarrhoea (Fig. 2a). Moving from no stagnant water to 3% of the surrounding buffer under water raises two-week diarrhoea prevalence by 4.83 percentage points ($t = 2.81$), relative to a sample mean of 10.7%. The response is monotonic over the range containing most observations and is well approximated by a smooth dose–response. Alternative functional forms yield similar estimates (Extended Data Fig. 6).

The response is considerably larger among children in households that rely on surface water for drinking (Fig. 2b–c). At 3% stagnant-water exposure, the difference in the response between surface-water drinkers and other households is +5.93 percentage points ($t = 2.04$). The two groups are statistically indistinguishable at zero stagnant-water exposure (−0.02 pp, $t = -0.02$), so the divergence arises in the response to stagnant water rather than in baseline diarrhoeal risk. This pattern is consistent with an enteric pathway in which stagnant surface water increases opportunities for faecal–oral exposure among households whose drinking water is drawn directly from the surrounding environment.

Because reliance on surface water also differs systematically across places, we test whether this heterogeneity can instead be explained by correlated household or local characteristics. Allowing the stagnant-water response to vary with poverty, sanitation, distance to water, maternal education and population density, separately and jointly, leaves the surface-water interaction essentially unchanged: the corresponding effect is +2.96 pp without these interactions and +3.16 pp when all five are included. Moreover, these characteristics are only weakly correlated with surface-water reliance ($r = 0.06$–0.11). Thus, the larger response among households drinking surface water is not readily

explained by these observable differences (Methods).

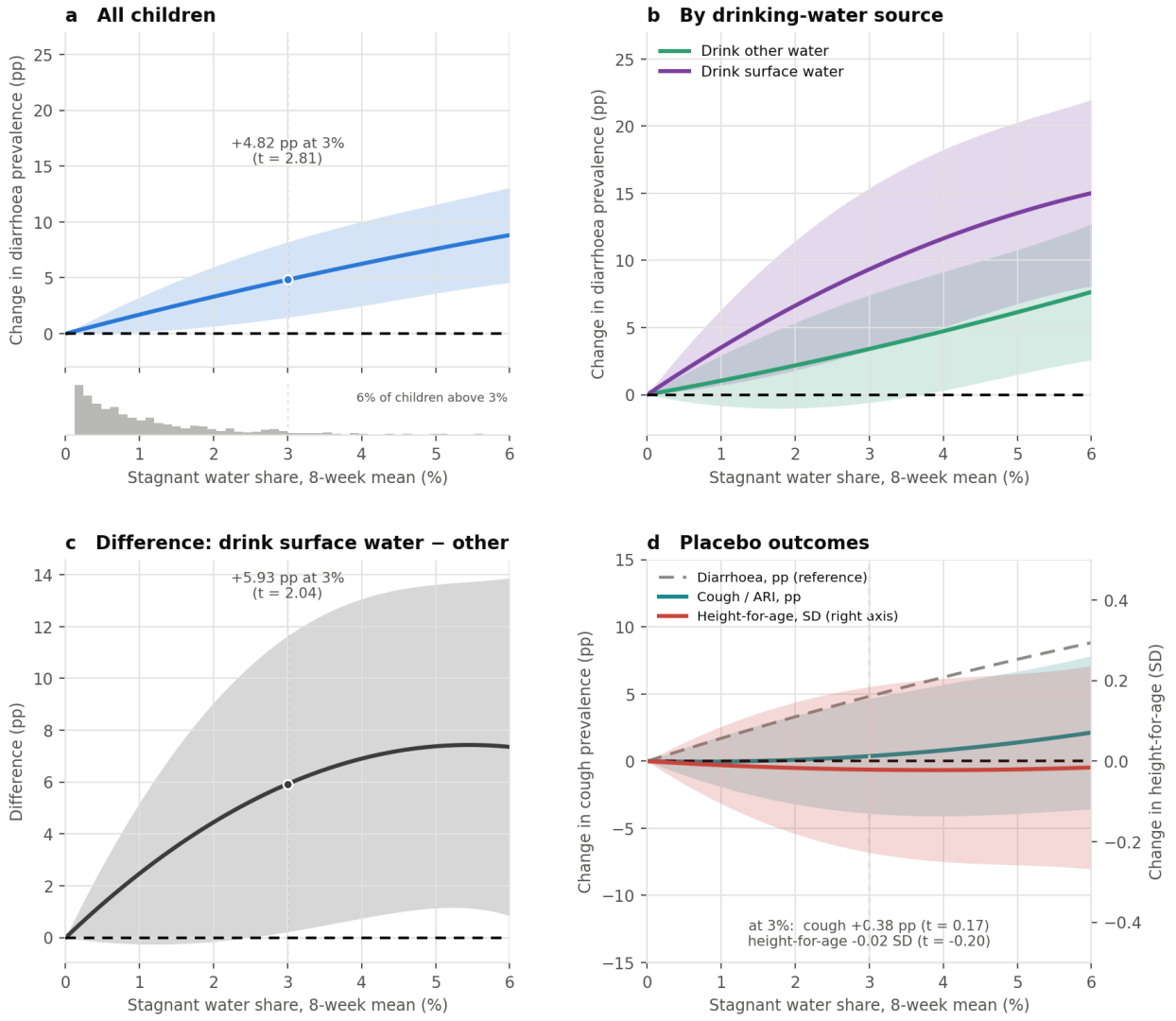


Fig. 2: Diarrhoea response to stagnant water among children under five. a, Main effect. **b**, Response by household drinking-water source among rural households. **c**, Difference in the dose–response between households relying on surface water and those using other drinking-water sources. **d**, Negative-control outcomes: cough and height-for-age, standardized to their own outcome distributions, with the diarrhoea response shown as a dashed reference. All curves show changes relative to zero stagnant-water exposure within the corresponding group; shaded areas denote 95% confidence intervals and standard errors are clustered at the DHS cluster level.

Stagnant water reduces haemoglobin in malaria-suitable areas

Stagnant water can also create breeding habitat for *Anopheles* mosquitoes. Malaria infection, in turn, lowers haemoglobin by destroying infected and uninfected red blood cells and suppressing erythro-

poietic recovery, making anaemia one of its most characteristic haematological consequences[5,40]. This implies a distinct biological signature: if stagnant water increases malaria exposure, reductions in haemoglobin should be concentrated where *Plasmodium falciparum* transmission is ecologically possible rather than where households rely on surface water for drinking. We therefore examine haemoglobin, an objective biomarker measured in 13,279 children, and stratify the response by malaria suitability using a threshold from the malaria-mapping literature[4,41].

Pooling across malaria suitability, moving from zero to 3% stagnant-water exposure reduces haemoglobin by 2.41 g/L ($t = -2.20$) (Fig. 3a). The pooled effect conceals a much sharper ecological pattern. In malaria-suitable areas haemoglobin falls steeply with stagnant-water exposure, whereas in unsuitable areas it does not. At 3% exposure, the difference between the two responses is -7.54 g/L ($t = -3.03$) (Fig. 3b–c).

The magnitude is epidemiologically meaningful. Malaria reduces haemoglobin through both haemolysis of infected and uninfected erythrocytes and impaired erythropoietic recovery[5]. In a quasi-experimental twin fixed-effects study of children under five across 23 sub-Saharan African countries, malaria infection was associated with a 9 g/L reduction in haemoglobin[40], placing the stagnant-water differential observed here in the same order of magnitude. This comparison should not be interpreted as implying that malaria alone accounts for the full haemoglobin response: our data measure ecological suitability rather than individual malaria infection, and stagnant water may affect haemoglobin through other water-related infections or short-run morbidity as well. Schistosomiasis, for example, is also associated with reduced haemoglobin[42].

Two additional outcomes nevertheless point specifically toward a vector pathway. Reported fever rises with stagnant water in malaria-suitable but not malaria-unsuitable areas, with a $+13.3$ percentage-point difference at 3% exposure ($t = 2.51$), and the probability of crossing the clinical anaemia threshold rises by 25.9 percentage points ($t = 3.67$) (Extended Data Fig. 2). Haemoglobin is our preferred main outcome because it is objectively measured rather than recalled and preserves the continuous variation underlying the anaemia classification.

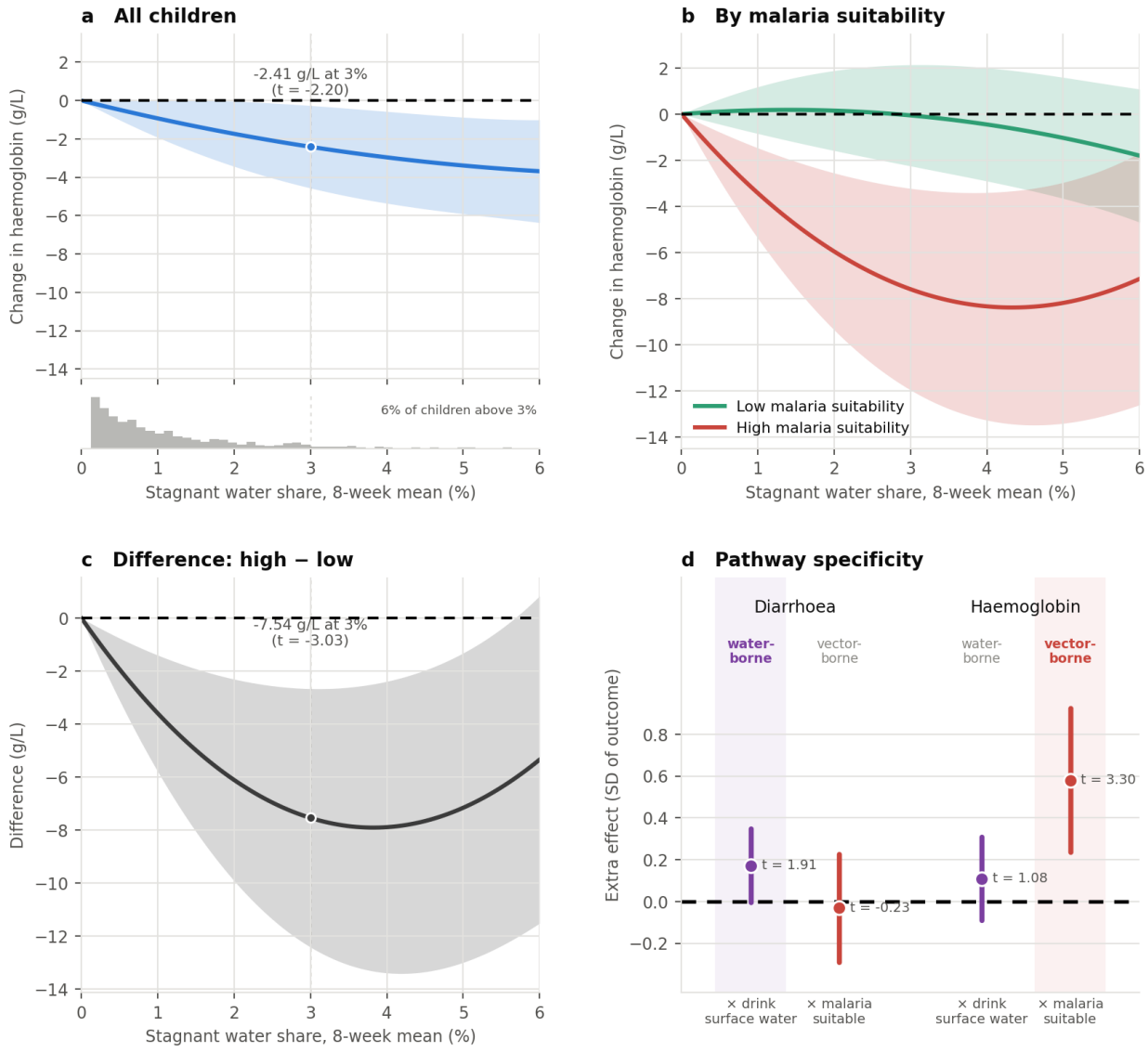


Fig. 3: Haemoglobin response to stagnant water is concentrated in malaria-suitable areas. **a**, Main effect. **b**, Response by malaria suitability. **c**, Difference between high- and low-suitability areas. **d**, Pathway specificity. Diarrhoea is more responsive to stagnant water among households relying on surface drinking water but not in malaria-suitable areas; haemoglobin shows the converse pattern. Effects in **d** are standardized to permit comparison across outcomes. Shaded areas denote 95% confidence intervals and standard errors are clustered at the DHS cluster level.

Pathway specificity and robustness

The two health responses, diarrhoea and haemoglobin loss, separate along the pathways predicted by their underlying biology (Fig. 3d). We estimate both moderators jointly for each outcome, so that the drinking-water interaction is conditional on malaria suitability and vice versa. Diarrhoea responds

more strongly when households rely on surface drinking water (+0.172 SD, $t = 1.91$), but not with malaria suitability (−0.031 SD, $t = -0.23$), suggesting an enteric pathway. Haemoglobin shows the opposite pattern: its response varies strongly with malaria suitability (+0.581 SD, $t = 3.30$) but not with drinking-water source (+0.110 SD, $t = 1.08$). The difference between these two pathway interactions is significant for haemoglobin (−0.471 SD, $t = -2.39$) and has the predicted sign but is less precisely estimated for diarrhoea (+0.203 SD, $t = 1.32$). A common seasonal shock or a general association between wet environments and poor health would be unlikely to generate this cross-outcome dissociation.

Placebo tests provide a complementary check (Fig. 2d). Cough, which has neither an enteric nor vector-borne transmission pathway, shows no response to stagnant water (+0.38 pp at 3%, $t = 0.17$). Height-for-age likewise does not respond (−0.02 SD, $t = -0.20$). Because height integrates nutritional and disease conditions over much longer periods than the eight-week exposure window, this null is informative in two ways: it argues against the estimates simply capturing persistent differences in child health between chronically wet and dry locations, and against long-run malnutrition being the primary explanation for the short-run haemoglobin response.

Further biological patterns reinforce the separation between pathways. Alternative malaria-suitability terciles produce a monotonic haemoglobin gradient, with progressively larger reductions as suitability rises, while diarrhoea shows a greater response in the lowest malaria suitability tercile (Extended Data Fig. 3). The outcomes also follow different age profiles (Extended Data Fig. 4). The diarrhoeal effect is weakest in infancy and peaks when children begin complementary feeding, drinking water and moving independently, whereas the haemoglobin profile is distinct and substantially less precise because of the smaller biomarker sample.

The estimates are also insensitive to the principal modelling and identification choices. Reconstructing exposure over windows ranging from two to twelve weeks leaves both effects similar (Extended Data Fig. 5); alternative polynomial and spline dose–response functions remain within the confidence band of the preferred specification (Extended Data Fig. 6). Adding rainfall, long-run flood proneness and household characteristics sequentially leaves the estimated responses essentially unchanged (Extended Data Fig. 7). Inference is similarly robust to allowing residual spatial correlation: replacing cluster-level standard errors with Conley spatial HAC estimates at 25, 50 and 100 km produces t-statistics of 2.63–3.07 for diarrhoea and −3.02 to −3.39 for the haemoglobin–malaria interaction (Extended Data Fig. 8).

The burden is concentrated in space and time

Combining the stagnant-water exposure surface of Fig. 1 with the estimated diarrhoea dose–response in Fig. 2 allows us to map the associated health burden across Tanzania. We evaluate the fitted

dose–response at each 2 km cell's own stagnant-water exposure and weight by the local under-five population. On this basis, stagnant water accounts for 11.3% (95% CI 4.3–18.7%) of childhood diarrhoea nationwide, equivalent to approximately 445 under-five deaths and 40,900 disability-adjusted life years per year at current mortality rates.

This conversion requires few modelling assumptions, since the national mortality burden is simply the estimated attributable fraction multiplied by the current number of under-five diarrhoeal deaths: $0.113 \times 3{,}924$. Population size, the number of two-week periods in a year and the implied case-fatality rate cancel algebraically (Methods). The resulting burden is therefore driven primarily by two quantities: the estimated exposure–response relationship and the underlying national diarrhoeal mortality rate.

That second quantity has changed markedly over time. Under-five diarrhoeal mortality in Tanzania declined by 84% between 2000 and 2023[43]. Applying the same estimated attributable fraction to the 2010 mortality rate, but today's under-five population, would imply 1,258 deaths per year, while applying it to the 2000 rate would imply 2,798. We report the burden as of 2023 throughout. These differences reflect the large decline in the probability that a diarrhoeal episode becomes fatal—through oral rehydration, vaccination and improved case management—rather than any assumed decline in stagnant-water exposure itself. Moreover, these calculations include only the diarrhoeal pathway in Fig. 2 and therefore exclude the vector-associated health effects identified in Fig. 3.

Concentrated in space. The burden is highly uneven geographically (Fig. 4a). The largest predicted burdens trace the same floodplains, inland deltas and low-relief basins in which the hydrological model of Fig. 1 predicts persistent accumulation of stagnant water. This spatial concentration carries directly into the distribution of health burden across children (Fig. 4c). The most exposed 5% of children account for 34% of the attributable burden, the most exposed 10% for 50%, and the most exposed 20% for 70%—equivalent to 153, 224 and 309 attributable deaths per year, respectively.

This concentration is not driven by a handful of sparsely inhabited locations. Approximately 1.07 million children under five—around one in ten nationally—live in cells where annual mean stagnant-water exposure exceeds 2% of the surrounding area, and these children account for half of the estimated national burden. Attributable mortality rises steeply across the exposure distribution, from approximately 0.3 deaths per 100,000 under-fives per year in cells with almost no stagnant water to 36 per 100,000 in cells where more than 5% of the surrounding area is under water. The concentration pattern is also robust to uncertainty in the estimated dose–response: even at the fifth percentile of the bootstrap distribution, the most exposed decile still carries 40% of the attributable burden, compared with 10% under an even distribution.

Concentrated in time. The same concentration appears over the calendar (Fig. 4d). The three months with the largest national burden account for 53% of the annual total, whereas the three quietest months account for only 2%; February alone accounts for 19%. Thus, the health burden generated by stagnant water is not only geographically concentrated but strongly seasonal.

The timing of that season, however, differs across the country (Fig. 4b). No single calendar month is the month of peak exposure for more than 38% of children. Exposure peaks in January and February across much of the western floodplains and the Rufiji and Kilombero systems, but later—often in May or beyond—along the eastern coast and on the islands. A national calendar therefore does not map cleanly onto local risk.

This joint concentration in space and time is what makes targeting possible. The same national intervention can reach very different amounts of health burden depending on whether it is directed towards the places and seasons in which stagnant water is concentrated. We turn to that allocation problem in Fig. 5.

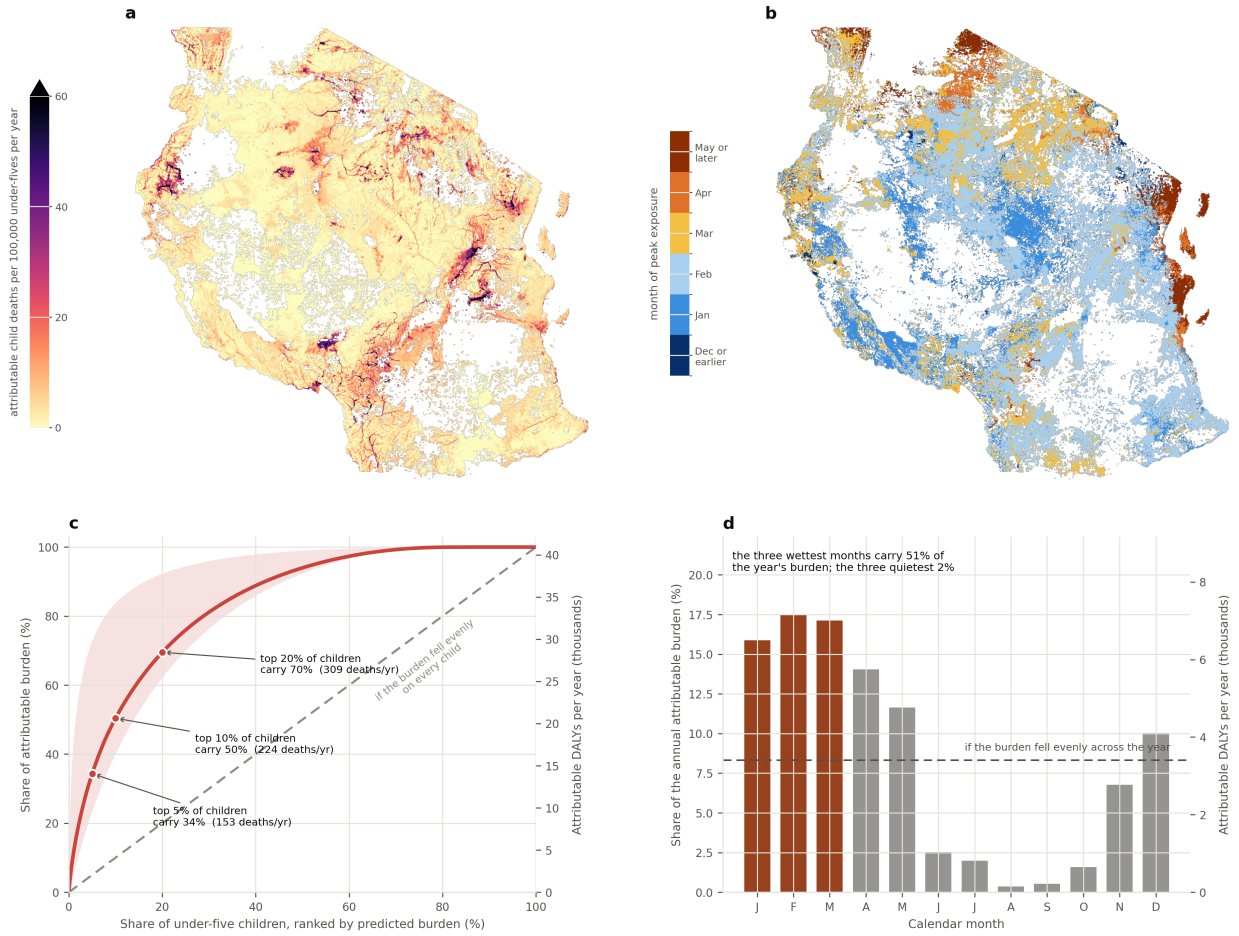


Fig. 4: The burden of stagnant water is concentrated in space and time. a, Estimated annual under-five diarrhoeal deaths attributable to stagnant water per 100,000 children on a 2 km grid, obtained by applying the dose–response from Fig. 2 to each cell's modelled exposure and weighting by the local under-five population. **b**, Calendar month of peak modelled stagnant-water exposure. Cells with peak exposure below 0.25% of surrounding area are omitted. **c**, Cumulative share of the national attributable diarrhoeal burden against the cumulative share of under-five children ranked by predicted burden. The dashed diagonal denotes an even distribution of burden. The shaded band is a 90% parametric-bootstrap interval over the estimated dose–response (300 draws), with cells re-ranked in each draw. **d**, Share of annual attributable burden occurring in each calendar month. Population weighting is used throughout.

Targeting gains from oral rehydration coverage

To illustrate how the stagnant-water estimates can inform existing health interventions, we consider a hypothetical policy effort to increase access to and uptake of oral rehydration therapy (ORS). How much greater would the resulting health gains be if the additional coverage were targeted using predicted stagnant-water risk over space and time?

ORS is among the most effective and inexpensive treatments for childhood diarrhoeal mortality, yet

coverage in Tanzania remains incomplete. In the DHS data, 45.2% of children with diarrhoea receive oral rehydration, with little change across the available survey waves (45.1% in 2010 and 43.8% in 2016), despite awareness above 96% (Extended Data Fig. 9). While the causes of incomplete coverage—including stockouts, provider behaviour, distribution and household uptake—are outside the scope of this paper, the relevant implication is that the scarcity of coverage means that targeting that maximizes health benefits is of particular importance.

Current ORS use shows little evidence of being targeted towards the burden identified by the stagnant-water model. Coverage is approximately flat across stagnant-water exposure and across household drinking-water source (Extended Data Fig. 9). We therefore treat the current allocation as an approximately untargeted benchmark and ask how many deaths could be averted if a small expansion in coverage were instead directed using information on stagnant-water risk.

A one-percentage-point increase in national ORS coverage corresponds to reaching an additional 1.8% of children and bringing those children to full coverage, which averts 88% of their diarrhoeal mortality risk (Methods). If this expansion is allocated in proportion to current coverage, it averts 69 under-five deaths per year. If it is instead targeted using the modelled stagnant-water exposure, it averts 139 deaths—approximately twice as many lives for the same increase in national coverage (Fig. 5a). The relative gain is larger for smaller expansions, because the burden is concentrated in the highest-exposure locations.

The comparison in Fig. 5a also shows how the value of targeting rises with the information available. A static annual map of stagnant-water exposure averts 118 deaths; adding location-specific seasonality raises this to 131; and using the current modelled water state raises it further to 139. By contrast, targeting with JRC Global Surface Water averts 66 deaths, slightly below the untargeted benchmark. This gap links the measurement result of Fig. 1 directly to the policy exercise: information on permanent and readily visible surface water is poorly aligned with the transient stagnant-water burden that drives the health effects estimated here.

These comparisons are scored on *all-cause* childhood diarrhoeal mortality, because ORS reduces the risk of death from diarrhoea regardless of its source. The targeting gain therefore comes from directing treatment towards places where stagnant-water exposure identifies a higher underlying diarrhoeal burden. If the objective were instead to target an intervention acting directly on stagnant water itself—for example drainage or larvicide interventions in stagnant water pools—the relevant outcome would be the burden attributable specifically to stagnant water. For this purpose, the same spatial targeting rule would instead be approximately 13 times as much burden as random allocation, making the gains in the exercise here closer to a lower bound.

How much lead time is needed? Operational targeting requires decisions to be made before exposure is fully realized. We therefore ask how much of the targeting gain survives when the model is used with increasing lead time (Fig. 5b). We bound performance using two extremes. The

lower bound assumes no rainfall-forecast skill and carries forward only the modelled water state observed L days earlier. The upper bound uses the same earlier water state together with the rainfall that subsequently occurred, equivalent to a perfect weather forecast over the lead window.

The two bounds remain close over operationally relevant horizons. At a five-day lead, persistence alone performs essentially as well as contemporaneous information, because the eight-week exposure measure changes little over such a short interval. Even at a 30-day lead, the live-model rule still averts about 132 deaths, or 95% of the zero-lead value, and performs similarly to the location-specific seasonal calendar. Thus, most of the available targeting gain can be captured without a high-frequency real-time system: updating the model roughly monthly using recent rainfall would preserve nearly all of the benefit.

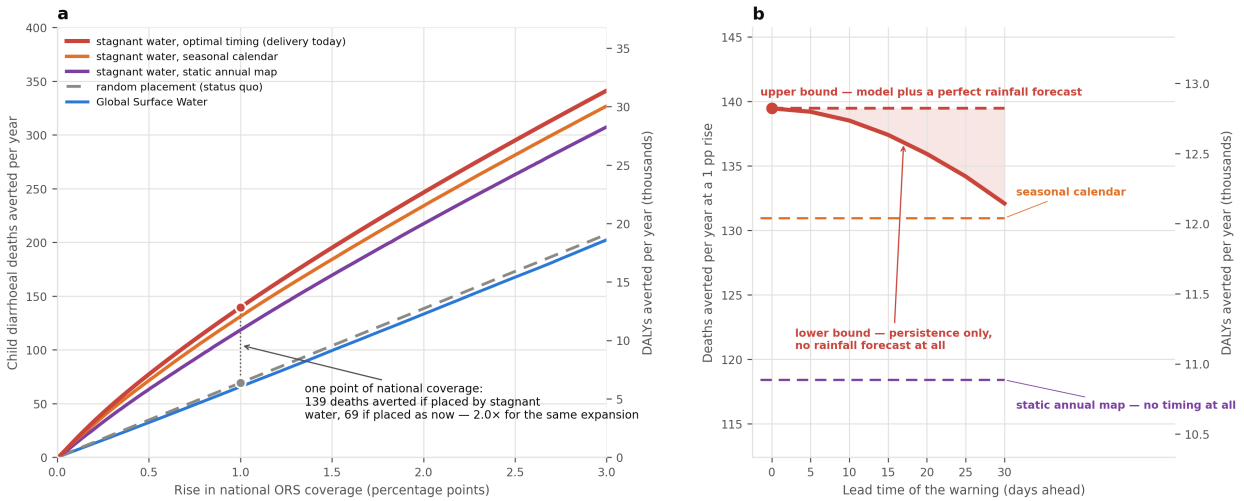


Fig. 5: Targeting additional oral rehydration coverage using stagnant-water risk. a, Estimated annual under-five diarrhoeal deaths averted by a marginal increase in national ORS coverage under alternative targeting rules. A one-percentage-point increase in national coverage corresponds to reaching an additional 1.8% of children and bringing them to full coverage; all rules are evaluated at the same total expansion. The right-hand axis gives the corresponding DALYs averted. **b**, Deaths averted under stagnant-water targeting as a function of decision lead time. The lower bound uses persistence of the previously modelled water state with no rainfall-forecast skill; the upper bound combines the previous state with subsequently realized rainfall, equivalent to a perfect forecast over the lead period. The horizontal reference denotes targeting using a static annual stagnant-water map. Exposure is the trailing eight-week mean used in Fig. 2. No intervention costs are assumed; comparisons therefore isolate the value of alternative targeting information at a fixed increase in coverage.

Two limitations bound this exercise. First, the ORS module is available only in the 2010 and 2016 surveys, so the data are not precise enough to rule out modest existing targeting towards high-risk areas. Under the main specification, the exposure gradient in ORS use is -8.9 pp at 3% exposure with a wide 90% interval of -36 to $+18$ pp ($t = -0.54$, $n = 1{,}398$), which suggests a negative relationship with stagnant water exposure, but does not reject the possibility of a positive

relationship.

Second, the location-specific seasonal calendar is estimated from only two to four model years per calendar month and is therefore noisy. If operationalized, additional model years would be expected to improve this benchmark and narrow its gap with the live model.

Discussion

This study identifies a widespread environmental exposure that is poorly represented in existing surface-water monitoring. Shallow, transient stagnant-water pools are difficult to observe with optical satellite products because they are short-lived, frequently vegetated and often form during cloudy wet-season conditions. Reconstructing these pools from rainfall, evaporation, soils and topography therefore provides a measure of surface-water exposure that is distinct from both conventional satellite products and rainfall itself.

Linking this exposure measure to nationally representative child-health data reveals two distinct health responses: an enteric response concentrated among households relying on surface drinking water, and a haemoglobin response concentrated where malaria transmission is ecologically possible. The broader implication is that moving from an upstream weather variable to the environmental state it generates can sharpen interpretation of climate–health relationships. Rainfall affects health not only through waterborne and vector-borne disease, but also through agricultural production, income, food security, infrastructure damage and other social and economic channels. A rainfall–health association can therefore be difficult to assign to any particular mechanism. By modelling where rainfall actually produces persistent surface water, the approach here isolates a more proximate exposure while retaining the large spatial and temporal coverage of gridded weather data[19–24].

This measurement gain has actionable policy implications with quantifiable returns. The attributable burden is highly concentrated across both places and seasons, so information on where stagnant water forms can be used to direct interventions rather than only to estimate average health effects. The oral-rehydration exercise illustrates this value for one existing intervention: conditional on achieving an increase in coverage, directing that increase using stagnant-water risk substantially raises the resulting health benefit. The same information could support source protection, drainage, vector control, surveillance or locally timed health-system preparedness, depending on the setting and transmission pathway.

Two considerations are important for interpreting these estimates. First, the exposure itself is modelled rather than directly observed. Validation against independent wetland and soil-moisture products establishes that the model reproduces plausible spatial and temporal hydrological variation, but not that every predicted pool exists at a particular place and date. Second, the absolute mortality burden is less certain than its spatial distribution. It combines the estimated attributable fraction

with an external estimate of current diarrhoeal mortality and therefore inherits uncertainty in both quantities. Tanzania's under-five diarrhoeal mortality has fallen sharply over the past two decades, so the same exposure distribution would have implied a substantially larger mortality burden when case management was weaker. By contrast, the concentration of burden and the relative gains from alternative targeting rules are less sensitive to the absolute mortality level.

The framework is readily portable beyond Tanzania because its physical inputs—elevation, soil properties, rainfall and evaporation—are globally available. The associated health response, however, need not be portable in the same way. Effects are likely to depend on local drinking-water systems, sanitation, vector ecology, health-care access and other environmental and institutional conditions. A natural extension is therefore to pair globally scalable stagnant-water prediction with large-scale health data and locally appropriate causal designs, so that exposure–response relationships and relevant transmission pathways are estimated in the same framework used to quantify burden and guide interventions. The analysis here illustrates this integrated approach: the same model is used to measure exposure, estimate pathway-specific health effects, quantify burden, and identify where and when intervention gains are greatest. Scaling the analysis across countries would also make it possible to study rarer but policy-relevant outcomes, including child mortality, for which the Tanzania DHS alone provides limited statistical power.

This approach may become increasingly useful as climate change alters the intensity, timing and episodicity of rainfall[44]. Forecast rainfall and climate projections can be propagated through the same hydrological model to identify not simply where rainfall changes, but where those changes generate persistent surface water. More broadly, the framework provides a way to translate weather into a more proximate environmental exposure—one that can be linked to health, mapped at fine scale and used for prediction and targeting.

Methods

Stagnant water model

Structure. We solve a water balance on the 90 m SRTM digital elevation model of Tanzania (EPSG:4326, $1/1200^\circ$). The state variable is water depth $h(x,t)$ above the land surface. Each hour the model (i) reads three-hourly rainfall and divides it evenly into five-minute increments; (ii) for each of twelve five-minute sub-steps, adds rainfall, redistributes water hydraulically, and removes infiltration; (iii) removes one hour of potential evapotranspiration; and (iv) floors depth at zero.

Hydraulic redistribution. For every cell with depth above a numerical threshold (10^{-6} m), we form the free-surface elevation

$$z = \mathrm{DEM} + h,$$

inspect the 3×3 neighbourhood, and identify the neighbouring cell with the lowest free-surface elevation. If the current level exceeds that minimum by more than floating-point tolerance, we transfer

$$\Delta \;=\; \min\!\left\{\, h,\; \tfrac{1}{2}\,(z - z_{\min}) \right\}$$

from the cell to that neighbour. The half-head factor relaxes adjacent water surfaces towards equilibrium while damping numerical oscillation. The scheme is explicitly mass-conserving, strictly local, and iterated once per five-minute sub-step, so water propagates at most one cell per sub-step. Boundary cells are held fixed.

The routing rule preserves the first-order hydraulic logic of standard shallow-flow approximations. Flow is directed down the local gradient in free-surface elevation, rather than bed elevation alone, so both terrain and dynamically varying water depth determine the hydraulic head difference between neighbouring cells. Larger head differences therefore produce larger transfers. In this respect the scheme resembles diffusive-wave routing, which retains the water-surface gradient, whereas the kinematic-wave approximation commonly closes the continuity equation using a uniform-flow relation such as Manning's equation[45,46].

Standard Manning-based routing additionally links flow velocity to hydraulic conveyance and surface roughness,

$$V \propto \frac{R^{2/3}}{n}\,|\nabla z|^{1/2},$$

where R is hydraulic radius and n is Manning's roughness coefficient[47,48]. Our rule instead makes transferred volume increase directly with the local difference in free-surface elevation. Water depth enters dynamically through $z = \mathrm{DEM} + h$ and through the constraint $\Delta \le h$, but the scheme contains no explicit roughness coefficient, friction law, sub-grid channel geometry or river cross-section.

This is a deliberate simplification, since the model is designed for nationwide application in a data-scarce setting, where spatially detailed roughness coefficients, channel cross-sections and other hydraulic parameters required for full kinematic-, diffusive- or dynamic-wave routing are not consistently observed. Introducing such inputs would therefore require strong spatial assumptions that are difficult to validate at the scale of an entire country. Our objective is not to reproduce channel discharge or flood-wave propagation, but to identify where rainfall-driven surface water is likely to accumulate and persist in low-relief terrain. For this purpose, a local mass-conserving hydraulic-head relaxation scheme provides a transparent compromise between physical structure, computational tractability and data requirements.

The model should therefore not be interpreted as a full shallow-water solver. It retains the dynamically varying free-surface gradient that governs the direction of redistribution, while abstracting from detailed momentum and frictional dynamics. Permanent water bodies are masked throughout and never contribute to predicted stagnant water.

Urban drainage. Soil data are unavailable for built-up land in our soil-data input[30]. Here, infiltration is assumed to be zero, since this surface is primarily paved. Instead built-up land is assigned a high effective drainage capacity, representing storm-water infrastructure. This helps to prevent artificial multi-week pooling on paved surfaces.

Summary of model inputs.

Input	Source	Native resolution	Temporal
Elevation	SRTM, mosaicked[29]	90 m	static
Soil saturated hydraulic conductivity	SoilGrids[30]	∼1 km	static
Permanent water mask	JRC Global Surface Water[16]	90 m	static
Rainfall	MSWEP v2[31]	0.1°	3-hourly
Potential evapotranspiration	ERA5-Land[49]	0.1°	hourly, accumulated

We use SRTM rather than the newer Copernicus DEM because COP90 is a digital *surface* model: it returns the top of the canopy and of buildings rather than the ground. For a model whose entire purpose is to route water across the land surface and find depressions, canopy height enters as spurious topography, and it does so most severely in exactly the vegetated settings where shallow ponding is hardest to observe from space.

Rainfall and evaporation are resampled to the DEM grid by nearest neighbour. Evaporation is supplied as an accumulated field and differenced to hourly increments. Missing rainfall and negative evaporation values are recoded to zero.

Runs and output processing. Each wave is simulated over a window beginning twelve weeks before the first interview date of that wave. The first four weeks are model spin-up, discarded, during which the water state equilibrates from a dry start; the following eight weeks constitute the exposure window for the earliest-interviewed clusters. The simulation then runs through the final interview date. Windows are wave-specific rather than continuous because each run takes several days of compute, which is the binding constraint on extending the model to a full multi-decade series.

For analysis we binarize at a depth of 5 cm, which we take as the threshold below which a pool is neither a meaningful mosquito habitat nor a plausible contamination source.

Exposure construction. For each DHS cluster we take the mean of the binary stagnant-water raster over a 2 km (urban) or 10 km (rural) buffer—matching the DHS displacement radii—and average over the eight weeks preceding the interview date[50]. When computing the spatial average exposure at each time step, we weight cells within each buffer by a triangular kernel, equal to one at the published coordinate and falling linearly to zero at the buffer edge (2 km urban, 10 km rural). For the 8-week temporal average, all weekly exposures are weighted equally. Extended Data Fig. 5 shows how estimates vary for alternative exposure windows, ranging from 2 to 12 weeks.

Validation

Wetland inventories. We compare model and GSW[16] coverage within Global Lakes and Wetlands Database classes[33], on an equal-area basis, using the 19 months in which model and GSW observations overlap. Basin-months are classified into rainfall quintiles using CHIRPS rainfall[51] averaged at the basin-month level, with basins defined as HydroBASINS level 7 grouped by Pfafstetter prefix[52].

Satellite soil moisture. We regress within-cell monthly anomalies in each water measure on within-cell anomalies in SMAP soil moisture (L3 AM retrieval and L4 assimilated product, $\sim$9 km), removing cell means from both sides, in 20 equal-count bins weighted by land area[39]. Both SMAP soil moisture measures yield similar levels of correlation, but only L3 AM retrieval is reported in Fig. 1.

Health data and sample

We use four waves of the Tanzania DHS (1999, 2010, 2015–16, 2021–22), restricted to children under five with non-missing outcomes and exposure, and to clusters where less than 20% of the buffer

has missing soil data. Outcomes are two-week recall diarrhoea, fever and cough. Haemoglobin is measured at the date of the interview and used to determine anaemia status, as is height. The analysis sample is 32,254 children; regression samples vary by outcome availability and are reported in each figure.

DHS coordinates are displaced before release—up to 2 km urban and 5 km rural, with 1% of rural clusters to 10 km—with displacement constrained to remain within the second administrative area[53]. We merge cluster coordinates from the DHS GPS files by survey cluster number within wave.

Estimation

The baseline specification is

$$y_{icgwm} = f(\mathrm{SW}_{cw}) + \gamma' X_{icgwm} + \alpha_{gw} + \lambda_{gm} + \varepsilon_{icgwm},$$

for child i in DHS cluster c, 0.5° grid cell g, survey wave w and calendar month m. SW_{cw} is the mean share of the surrounding buffer covered by stagnant water over the eight weeks preceding interview, and $f(\cdot)$ is quadratic in exposure. X includes rainfall over the same eight-week window, mean GSW occurrence in the buffer, and controls for wealth quintile, mother's age, urban status, child age and sex. The terms α_{gw} and λ_{gm} denote grid-cell×wave and grid-cell×calendar-month fixed effects, respectively. Standard errors are clustered at the DHS cluster level. In heterogeneity specifications, the full quadratic exposure function and the rainfall and GSW controls are interacted with the moderator.

The main role of the fixed effects is to enable local comparisons between DHS clusters while accounting for shared seasonality effects that might otherwise confound the results. Because DHS redraws survey clusters across waves, the same location is generally not observed repeatedly. Grid-cell×wave fixed effects therefore compare children in different clusters within the same 0.5° cell and survey wave, absorbing shocks and environmental conditions common to that local area at the time of the survey. Grid-cell×month fixed effects additionally absorb location-specific seasonality that recurs in the same calendar month across survey waves. Identification of the stagnant-water coefficient thus comes from fine-scale differences in exposure across nearby clusters observed under the same broad local and temporal conditions, rather than from comparisons across distant regions or survey waves.

Inference. Treatment is assigned at the DHS cluster-level, and in our baseline specification standard errors are correspondingly clustered there[54]. Since treatment may be correlated between DHS clusters, we also apply Conley spatial HAC standard errors at cutoffs of 25, 50 and 100 km which yield t-statistics between 2.63 and 3.07 for diarrhoea and between -3.02 and -3.39 for

haemoglobin (Extended Data Fig. 8). All three choices leave both results significant at conventional levels, and we report the DHS cluster-level one throughout.

Burden and policy calculation

Population-attributable burden. We first calculate the population attributable fraction (PAF), defined here as the share of observed diarrhoea prevalence attributable to stagnant-water exposure relative to a counterfactual of zero exposure. For each 2 km cell j, we evaluate the fitted dose–response at that cell's exposure, SW_j, and compute the predicted increment relative to zero exposure,

$$\Delta\hat{p}_j = \hat{p}(\mathrm{SW}_j) - \hat{p}(0).$$

Weighting cells by their under-five population N_j, the national PAF is

$$\mathrm{PAF} = \frac{\sum_j N_j \Delta\hat{p}_j}{\bar{p}_{\mathrm{obs}} \sum_j N_j},$$

where $\bar{p}_{\mathrm{obs}}$ is observed two-week diarrhoea prevalence in the DHS sample (10.7%). This yields a national attributable fraction of 11.3%.

From attributable fraction to mortality. Converting this fraction into deaths requires only one external mortality input: the national number of annual under-five diarrhoeal deaths. To see this, annual diarrhoea episodes can be written as $\bar{p}_{\mathrm{obs}} \times 26 \times N$, where N is the under-five population. If the episode case-fatality rate is calibrated as

$$\mathrm{CFR} = \frac{D}{\bar{p}_{\mathrm{obs}} \times 26 \times N},$$

where D denotes national annual diarrhoeal deaths, then attributable deaths are

$$(\mathrm{PAF} \times \bar{p}_{\mathrm{obs}} \times 26 \times N) \times \mathrm{CFR} = \mathrm{PAF} \times D.$$

Thus population size, observed prevalence and the number of two-week periods per year cancel algebraically; the mortality conversion depends only on the estimated PAF and the national diarrhoeal death count.

We take D from the Global Burden of Disease (GBD) 2023 estimates for Tanzania[43]: 36.0 diarrhoeal deaths per 100,000 children under five (95% uncertainty interval, 23.6–51.8). Applied to an under-five population of 10.9 million, this gives $D = 3{,}924$ annual deaths and therefore

$$0.113 \times 3{,}924 = 445$$

deaths attributable to stagnant water. The uncertainty interval in the GBD mortality rate implies 272–598 deaths, which we report separately from the 168–734 (95%) interval obtained by bootstrapping the estimated dose–response, because these capture different sources of uncertainty.

Disability-adjusted life years are converted at the ratio GBD reports for the same population, 3,310 DALYs per 100,000 against 36.0 deaths per 100,000, or 91.9 DALYs per diarrhoeal death. The population raster is GHS-POP for 2010, which we use for the spatial distribution only; the national under-five total is scaled to the year 2023.

The one crucial assumption of this exercise we cannot test is that a diarrhoea episode caused by stagnant water exposure is assumed to have the same case fatality as other episodes. If they are instead more severe, or if the places where they occur have worse access to care—both plausible, given that exposure is rural and concentrated in low-lying, poorly drained terrain—the true burden is larger and our figure is conservative.

Exposure in the placement exercise. The dose–response in Fig. 2 is estimated using mean stagnant-water exposure over the eight weeks preceding interview. To keep the policy exercise on the same exposure scale, Fig. 5 ranks cells using the corresponding trailing eight-week mean. At each five-day model snapshot, we therefore average the current and ten preceding snapshots (55 days in total), restricting the calculation to observations within the same model run.

The allocation problem is subject to a *population* budget. For a given increase in national coverage, the planner can treat a fixed number of children, equal to the coverage increment multiplied by the under-five population. Cells are therefore ranked by expected benefit *per child*, rather than by total burden in the cell, and are selected in priority order until the budget is exhausted. This is the appropriate greedy rule when the marginal cost scales with the number of children reached: a more populous cell contains more total burden, but also uses proportionally more of the available budget. Population therefore enters through the budget constraint rather than the ranking score. If instead programme costs were dominated by a fixed cost per delivery site or village, ranking by total cell burden would be more appropriate and would place greater weight on populous cells.

From additional coverage to deaths averted. ORS coverage among children with diarrhoea in these data is 45.2%, and is similar across survey waves (45.4% in 2010 and 45.0% in 2016, Extended Data Fig. 9). We therefore take baseline coverage to be $c_0 = 0.452$. This is a population coverage rate: a share c_0 of diarrhoeal episodes receive ORS, while the remainder do not.

Let e denote the reduction in diarrhoeal mortality among episodes receiving ORS. If mortality in the absence of treatment is normalized to one, expected mortality at coverage c is proportional to

$$1 - ec.$$

Increasing coverage in a targeted population from the current level c_0 to full coverage therefore averts the following share of *current* diarrhoeal mortality:

$$\frac{(1-ec_0)-(1-e)}{1-ec_0} = 1 - \frac{1-e}{1-ec_0}.$$

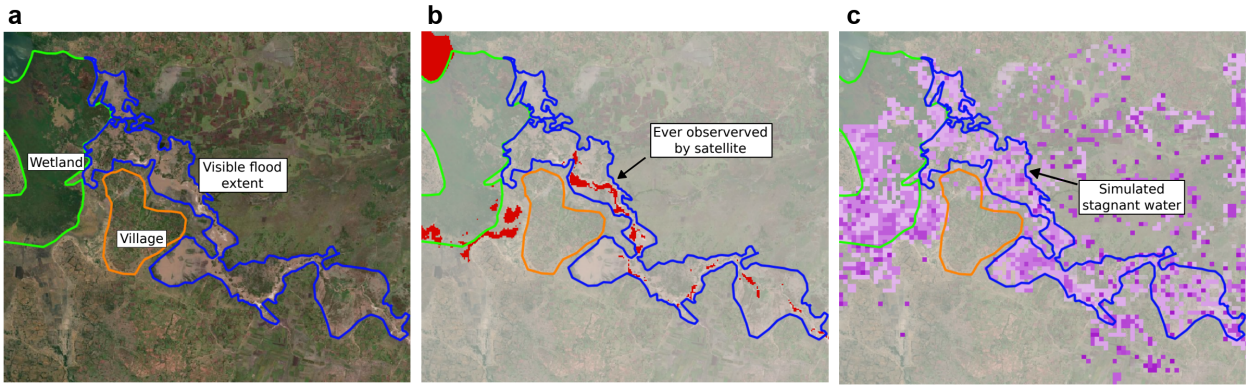
We set $e = 0.93$ following Munos et al.[55], who estimate that ORS at full coverage could prevent 93% of diarrhoeal deaths. At the observed baseline coverage $c_0 = 0.452$, moving a targeted population to full coverage therefore averts approximately 88% of its current diarrhoeal mortality risk.

This factor multiplies every placement rule identically. It determines the absolute number of deaths averted in Fig. 5, but cancels from comparisons between alternative targeting rules. Because the implied fraction is also only weakly sensitive to observed differences in baseline coverage across wealth groups, we hold c_0 constant across cells.

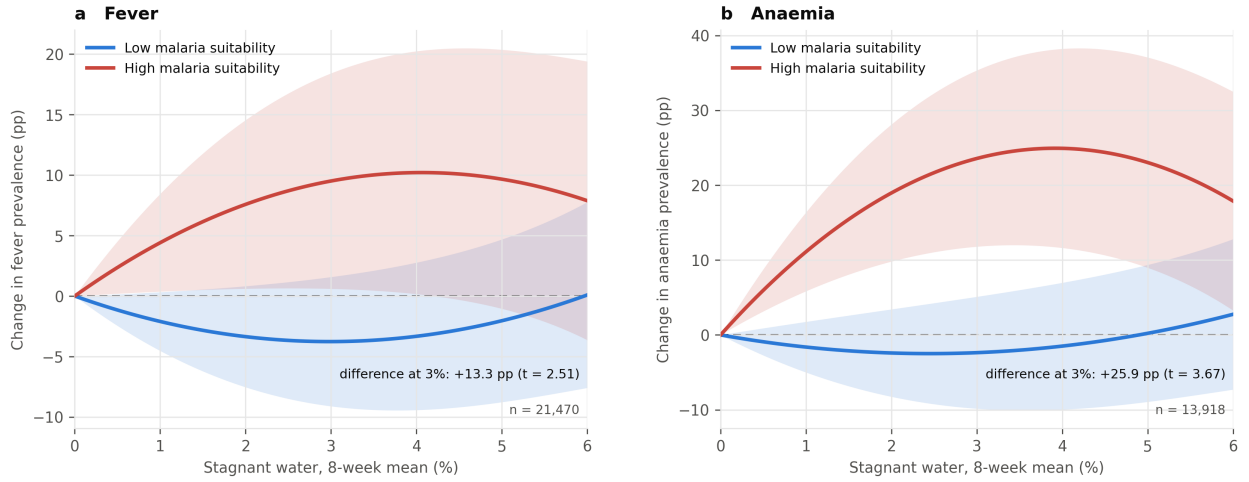
Code and data availability

To be added later.

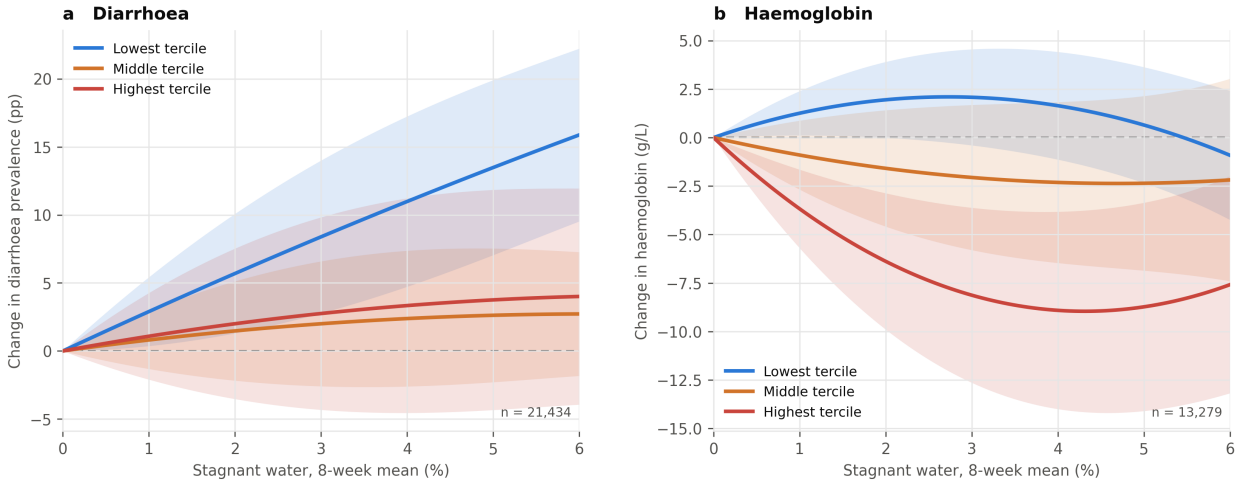
Extended Data



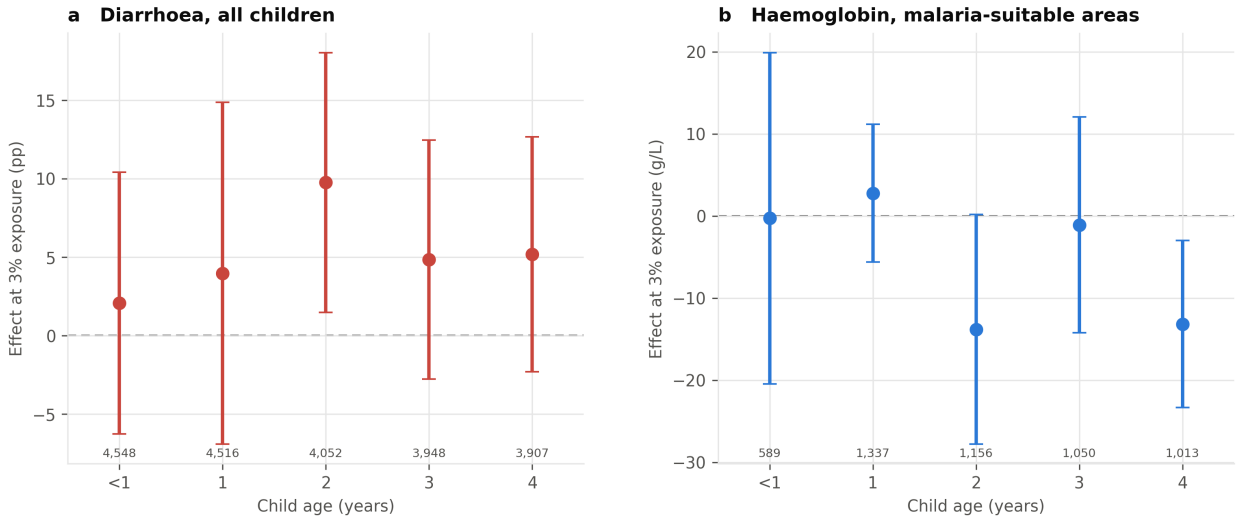
Extended Data Fig. 1 | Wet-season inundation around Muchirobi illustrates water missed by an optical surface-water product. The village of Muchirobi, Tanzania (approximately 1.4° S, 34° E; WGS84), during the wet season in May 2022. **a**, High-resolution satellite imagery indicates shallow brown pools of stagnant water across several agricultural plots within the flood extent outlined in blue. The village is outlined in orange, and a nearby wetland, where dense vegetation masks any surface water, in green. **b**, The same area with water marked as ever observed by the optical satellite Global Surface Water[16] in red. **c**, The same area with simulated stagnant water in purple.



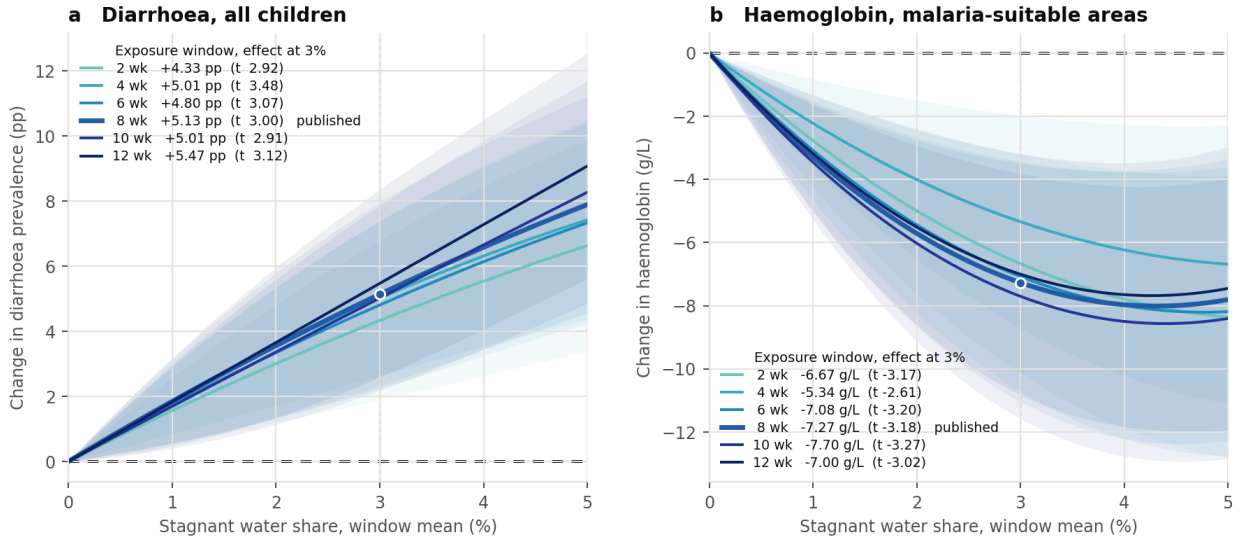
Extended Data Fig. 2 | Additional outcomes supporting the vector pathway. a, Fever and **b**, anaemia responses to stagnant-water exposure, estimated separately by malaria suitability under the specification of Fig. 3. Both outcomes respond more strongly to stagnant water in malaria-suitable areas. At 3% exposure, the difference between malaria-suitable and unsuitable areas is +13.3 percentage points (pp) for fever ($t = 2.51$, $n = 21{,}470$) and +25.9 pp for anaemia ($t = 3.67$, $n = 13{,}918$). Anaemia provides a clinically defined counterpart to the continuous haemoglobin outcome in Fig. 3, while fever provides a symptom-based outcome reported by the child's caregiver.



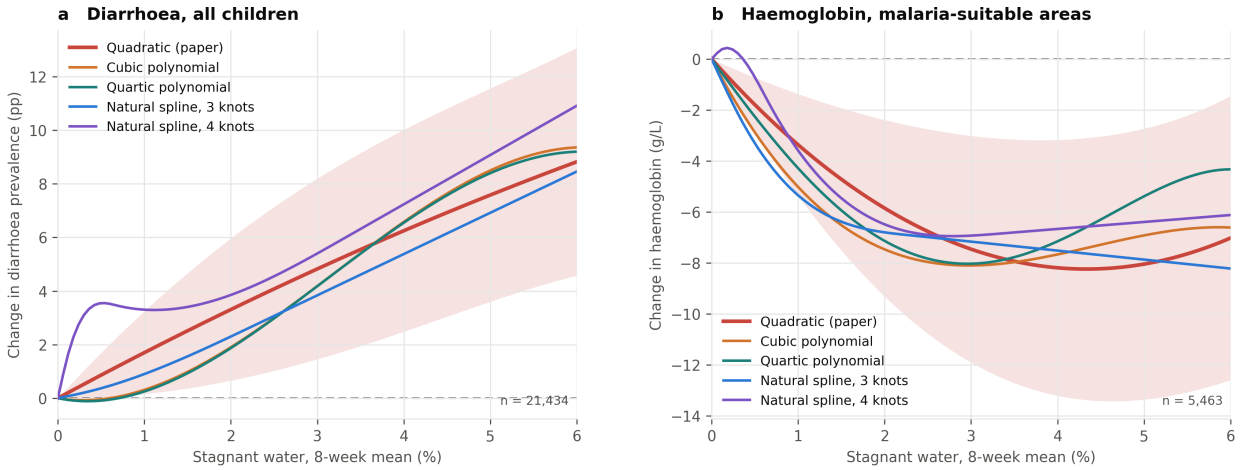
Extended Data Fig. 3 | Health responses across terciles of malaria suitability. **a**, Diarrhoea and **b**, haemoglobin responses to stagnant-water exposure across terciles of malaria suitability. Whereas Fig. 3 uses a single threshold from the malaria-mapping literature, terciles allow the response to vary across the suitability distribution. Relative to the lowest tercile, the haemoglobin response at 3% exposure becomes progressively more negative in the middle (-4.1 g/L, $t = -1.81$) and highest (-10.2 g/L, $t = -3.89$) terciles. Diarrhoea shows no corresponding monotonic gradient (-6.4 pp and -5.6 pp; $t = -1.73$ and -1.23). Tercile cutoffs are 0.240 and 0.429. Shaded bands show 95% confidence intervals.



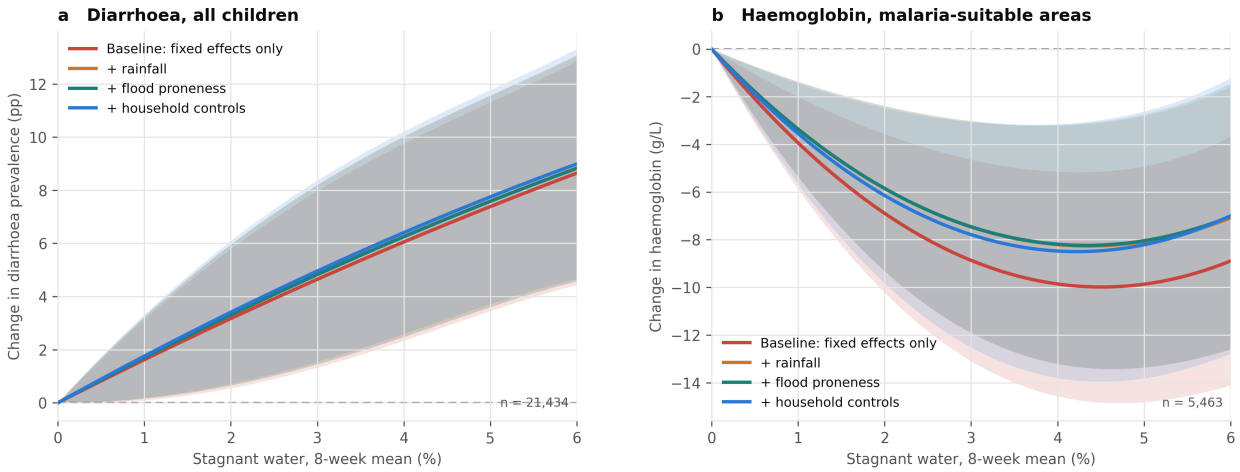
Extended Data Fig. 4 | Age profiles of the two health responses. Effects at 3% stagnant-water exposure estimated separately within each year of child age. **a**, Diarrhoea. The estimated effect is smallest in infancy ($+2.1$ pp, $t = 0.49$) and largest in the third year of life ($+9.8$ pp, $t = 2.31$), when children are increasingly mobile and exposed to complementary food and water. **b**, Haemoglobin. Estimates are shown separately by age using the malaria-suitability interaction in Fig. 3. The haemoglobin samples are smaller ($n = 589$–$1{,}337$ per age) and should therefore be interpreted with correspondingly greater uncertainty. Sample sizes are reported beneath each estimate. Error bars show 95% confidence intervals.



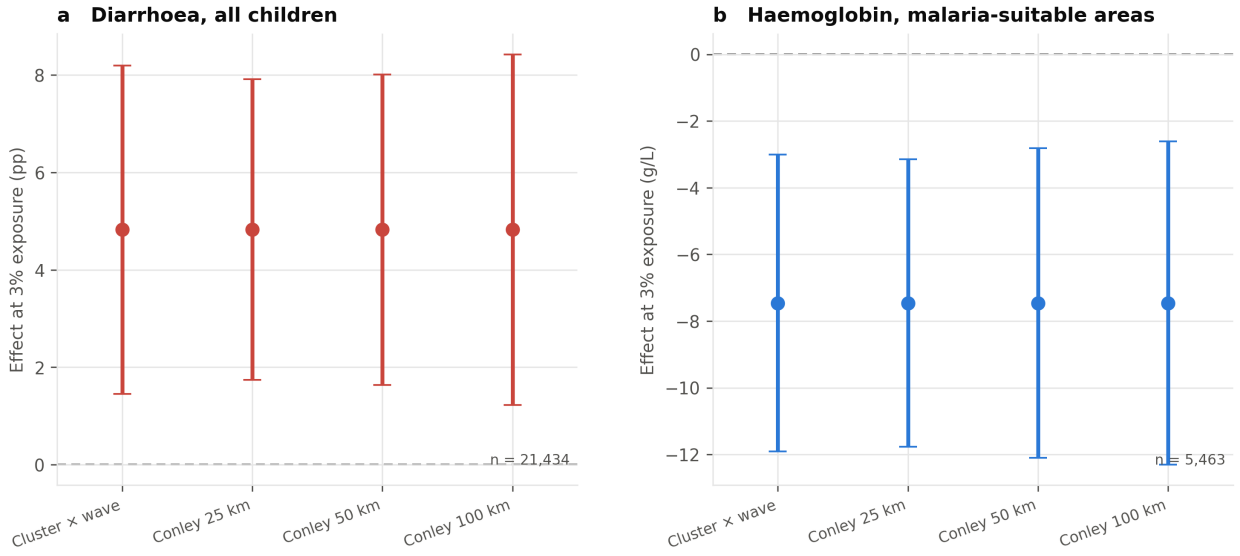
Extended Data Fig. 5 | Robustness to alternative exposure windows. a, Diarrhoea and **b**, haemoglobin (in malaria-suitable areas) effects estimated after reconstructing stagnant-water exposure over windows ranging from two to twelve weeks. All specifications use the triangular spatial kernel and a common sample of children for whom every exposure window is available. The diarrhoea effect at 3% exposure ranges from +4.3 pp at two weeks to +5.5 pp at twelve weeks ($t = 2.92$–3.12), while the haemoglobin effect ranges from −5.3 to −7.7 g/L ($|t| = 2.61$–3.27). For comparison, the corresponding eight-week estimates in the full analysis samples are +4.8 pp and −7.6 g/L. Rainfall controls are constructed over the corresponding exposure window wherever weekly data permit. Error bars show 95% confidence intervals.



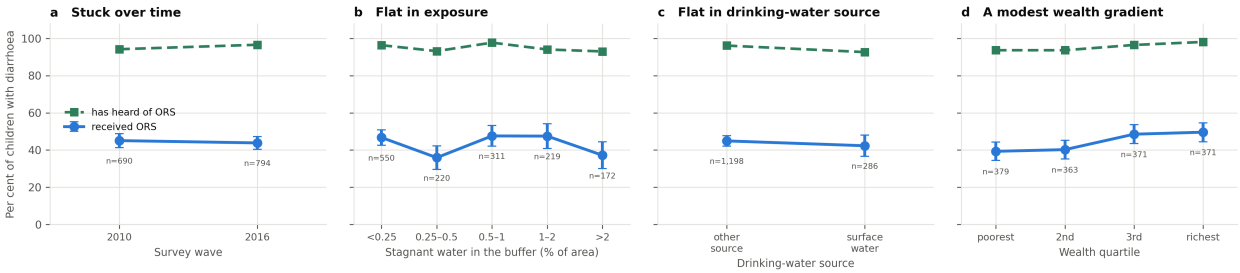
Extended Data Fig. 6 | Robustness to alternative functional forms. a, Diarrhoea and **b**, haemoglobin (in malaria-suitable areas) dose–response functions estimated using a quadratic, cubic and quartic polynomial and natural cubic splines with three and four knots. All specifications use the same sample and fixed effects and are normalized to zero at zero stagnant-water exposure. The alternative functions lie within the 95% confidence band of the baseline quadratic over the empirically relevant exposure range. At 3% exposure, the estimated diarrhoea effect ranges from +3.8 to +5.4 pp across specifications and the haemoglobin effect from −6.9 to −8.1 g/L. The shaded band is shown for the baseline quadratic only to avoid obscuring the comparison.



Extended Data Fig. 7 | Robustness to additional controls for persistent spatial differences. a, Diarrhoea and **b**, haemoglobin dose–response functions as additional controls are introduced sequentially on a common sample. The specifications progressively add controls intended to capture persistent differences between more- and less-exposed locations. Across the sequence, the diarrhoea effect at 3% exposure changes from +4.7 to +4.8, +4.8 and +5.0 pp, while the haemoglobin effect changes from −8.9 to −7.5, −7.5 and −7.8 g/L. The common samples contain 21,434 children for diarrhoea and 5,463 for haemoglobin. Flood proneness is measured by long-run Global Surface Water[16] occurrence within the exposure buffer. Additional household controls include maternal literacy, sanitation category, toilet type and drinking-water source, entered categorically with missing values retained as a separate category. Wealth quintile, urban status, maternal age, child age and sex are included in the baseline specification. Shaded bands show 95% confidence intervals.



Extended Data Fig. 8 | Robustness of statistical inference to spatial correlation. Comparison of baseline standard errors clustered at the DHS cluster level with Conley spatial heteroskedasticity- and autocorrelation-consistent standard errors using Bartlett kernels with 25, 50 and 100 km cutoffs. **a**, For diarrhoea, the standard error of the effect at 3% exposure is 1.72 under baseline clustering and 1.57, 1.63 and 1.84 under 25, 50 and 100 km Conley cutoffs, corresponding to $t = 2.81$, 3.07, 2.97 and 2.63. **b**, For haemoglobin, the corresponding standard errors are 2.27, 2.20, 2.37 and 2.47, corresponding to $t = -3.29$, -3.39, -3.15 and -3.02. Error bars show 95% confidence intervals.



Extended Data Fig. 9 | Oral rehydration coverage, awareness and distribution in the DHS. Children with diarrhoea in the two weeks before interview for whom the oral-rehydration module was completed ($n = 1{,}484$ across 431 clusters; the module is available in the 2010 and 2016 waves). **a**, ORS use and awareness by survey wave. Use remains near 45% while awareness rises from 94% to 97%, indicating substantial remaining coverage despite near-universal awareness. **b**, ORS use across stagnant-water exposure. **c**, ORS use by drinking-water source. Coverage varies little along either dimension, supporting the use of the observed allocation as an approximately uninformative benchmark with respect to the stagnant-water burden in Fig. 5. **d**, ORS use by wealth quartile. Error bars for coverage rates show 95% binomial confidence intervals.

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